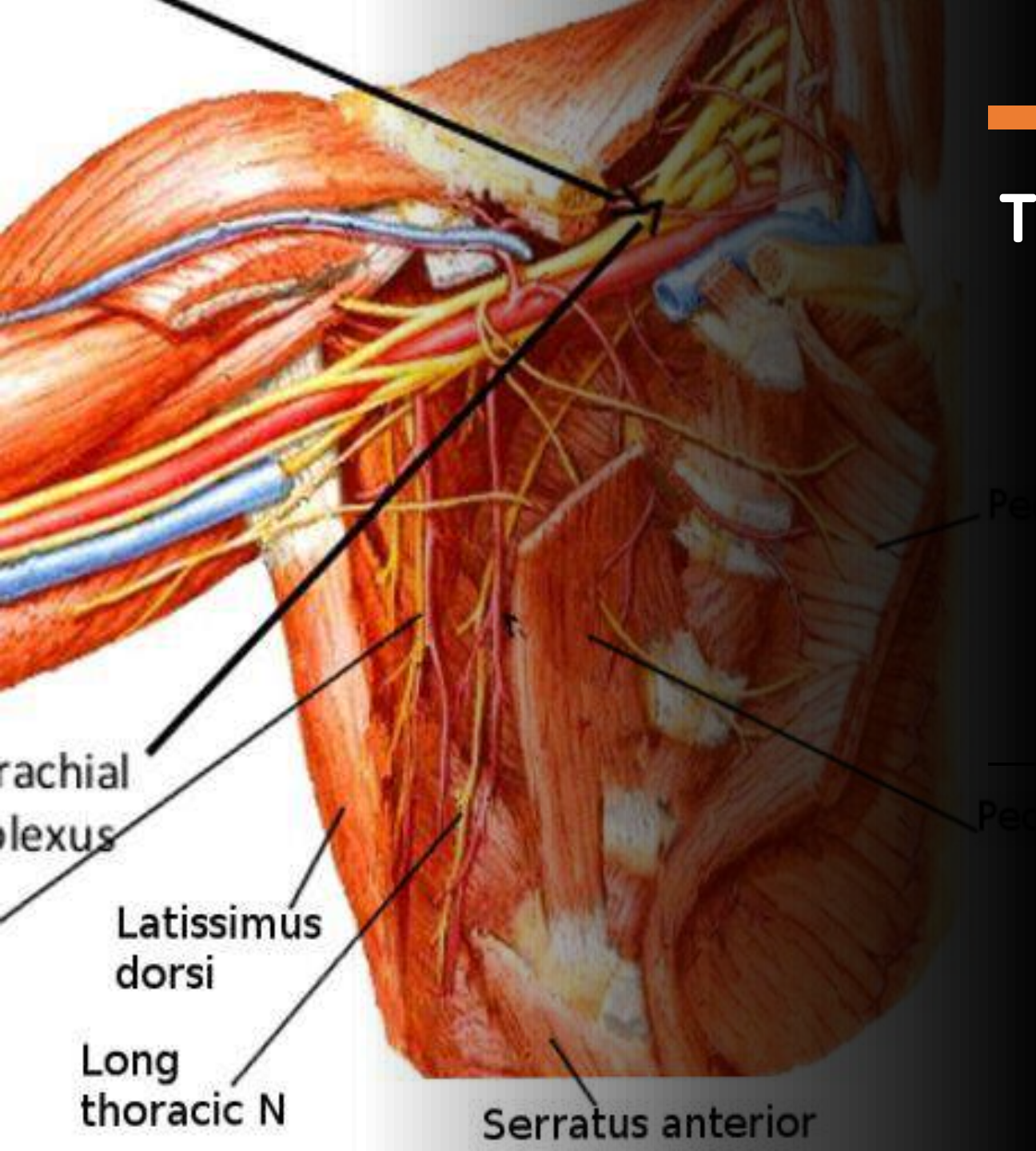


The Upper Limb

The Axilla



Dr. Rasha Alshali

OBJECTIVES



- ❑ The axilla: boundaries and contents.
 1. The **Axillary artery**:
 - ✓ Its course, branches, main relations
 - ✓ Its surface anatomy & site of pulsation.
 2. The **Axillary vein** the course & tributaries.
 3. The **Brachial plexus** & its branches.
 4. The **Axillary lymph nodes**

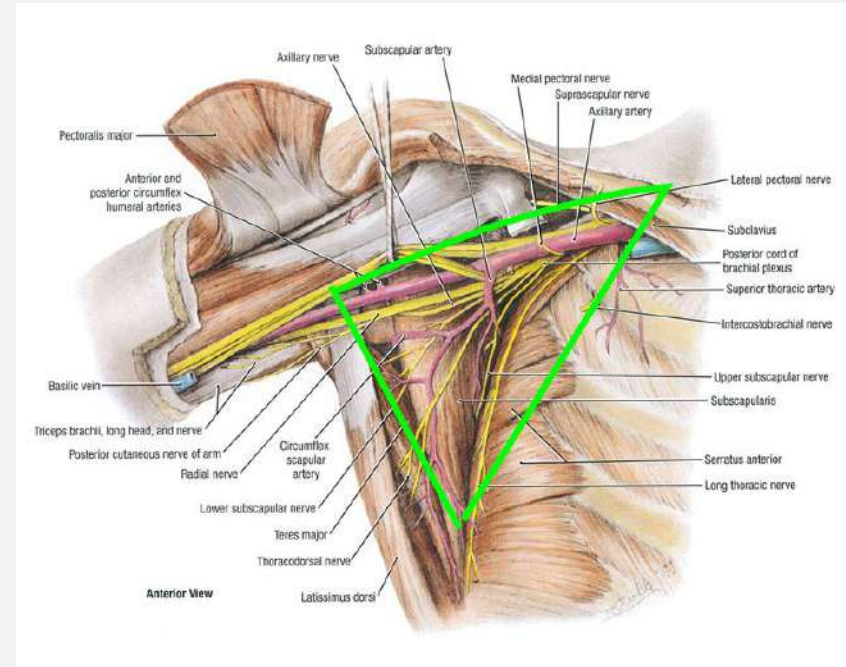
What is the Axilla ?

It is the **armpit**,

A pyramid-shaped space between the upper part of the arm and the side of the chest.

Clinically:

It is an important passage for nerves, blood vessels & lymph vessels from the **root of the neck to the upper limb**.

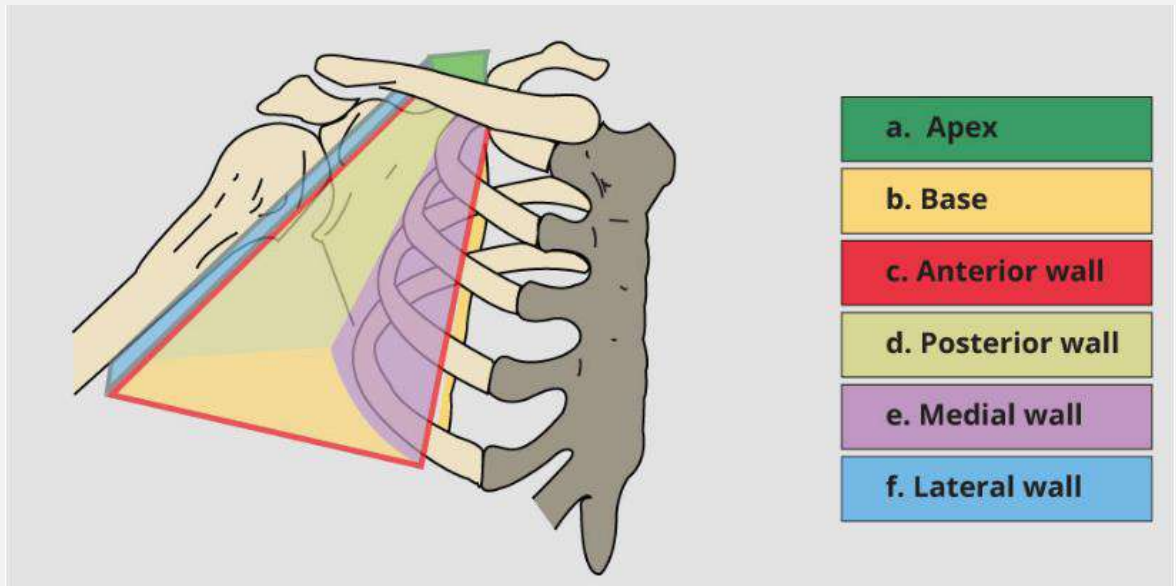


It has:

The **apex** (inlet)

The **base** (outlet)

The **walls**

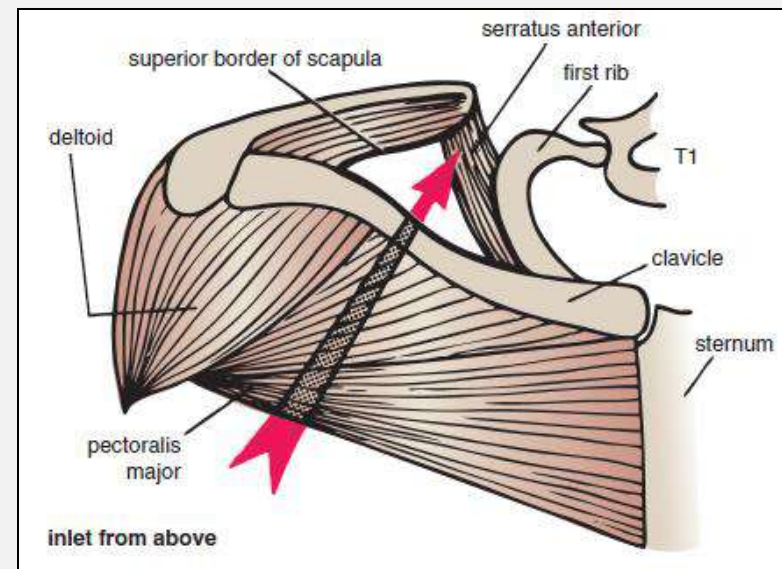


The apex: (cervico-axillary canal)

The upper end of the axilla, directed upward into the root of 'neck.

Bounded:

- Anteriorly by the **clavicle**,
- posteriorly by 'upper border of 'scapula,
- medially by 'outer border of 'first rib.

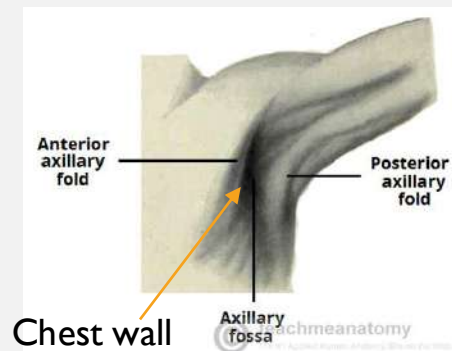
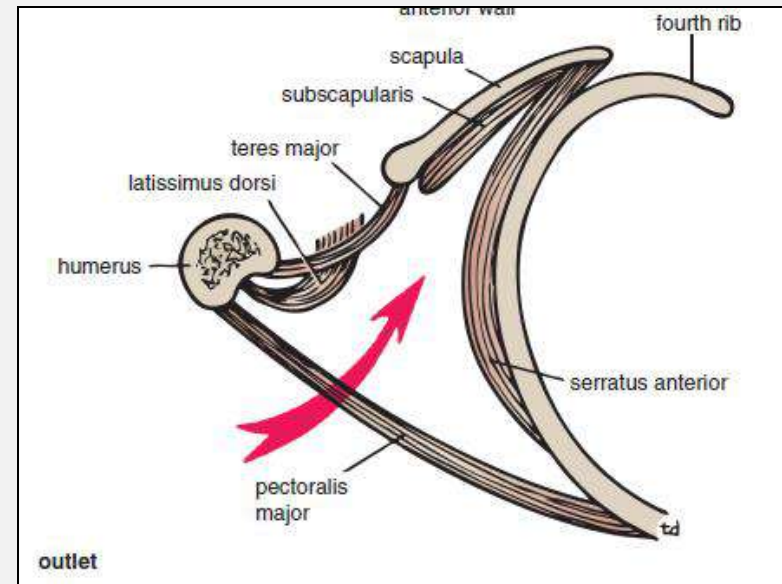


The base:

The lower end, formed by 'skin stretching between the anterior & posterior walls.

Bounded:

- medially by the **chest wall**.
- anteriorly by 'anterior axillary fold
- posteriorly by 'posterior axillary fold



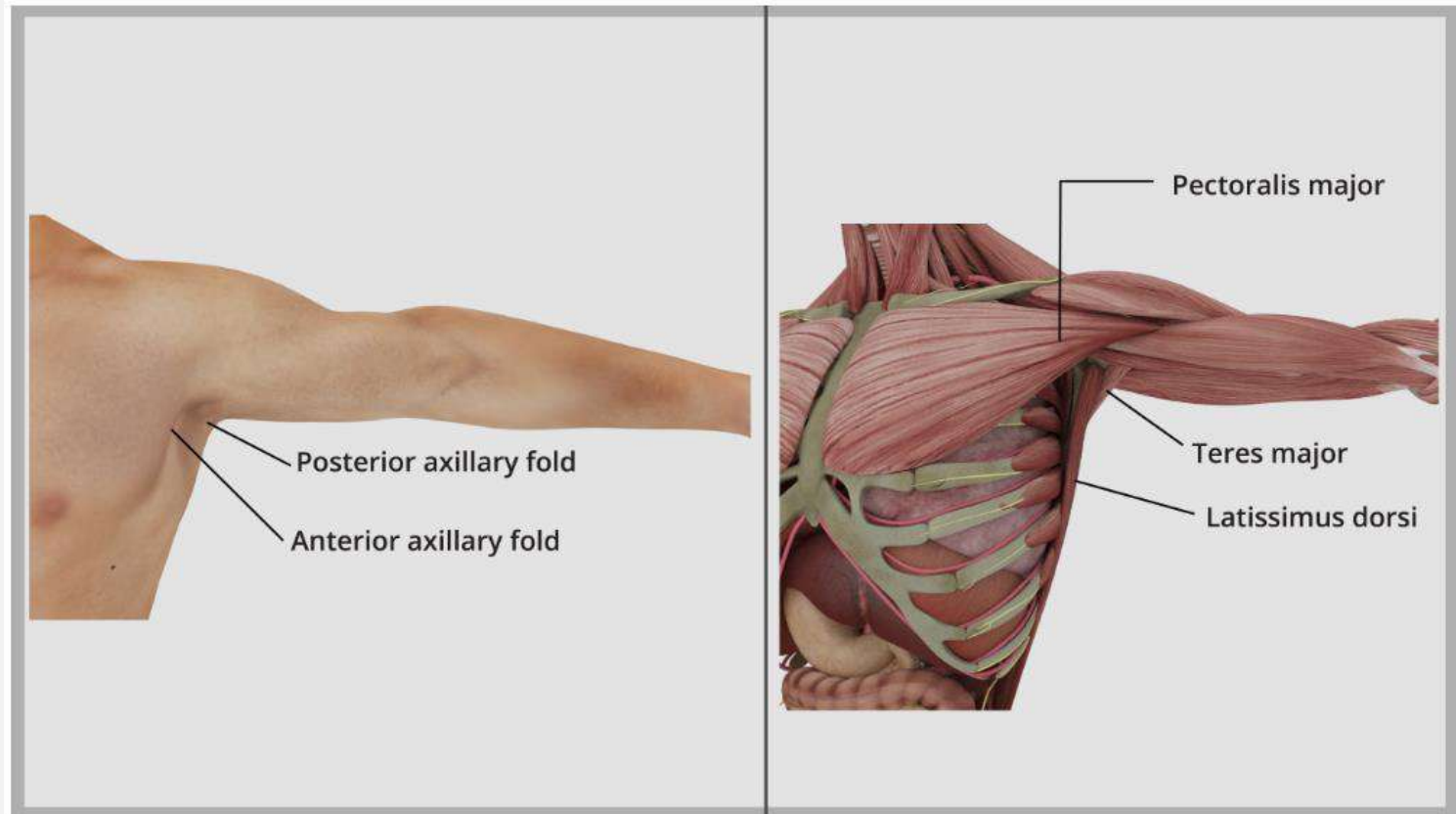
AXILLARY FOLDS

Anterior axillary fold

(formed by 'lower border of 'pectoralis major muscle)

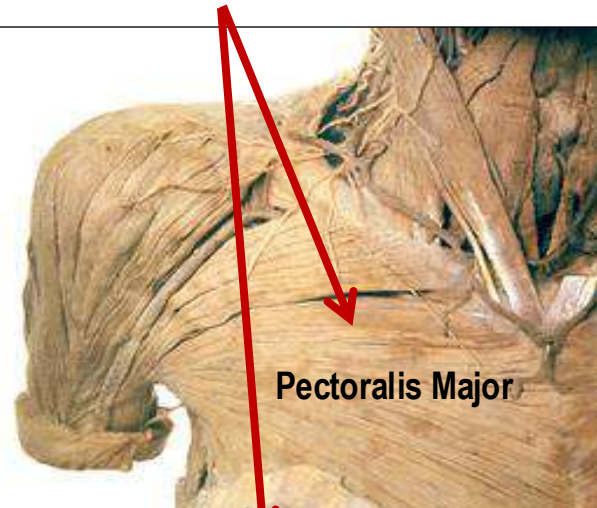
Posterior axillary fold

(formed by 'tendon of latissimus dorsi and the teres major muscles)



Walls of the Axilla

Anterior Wall:

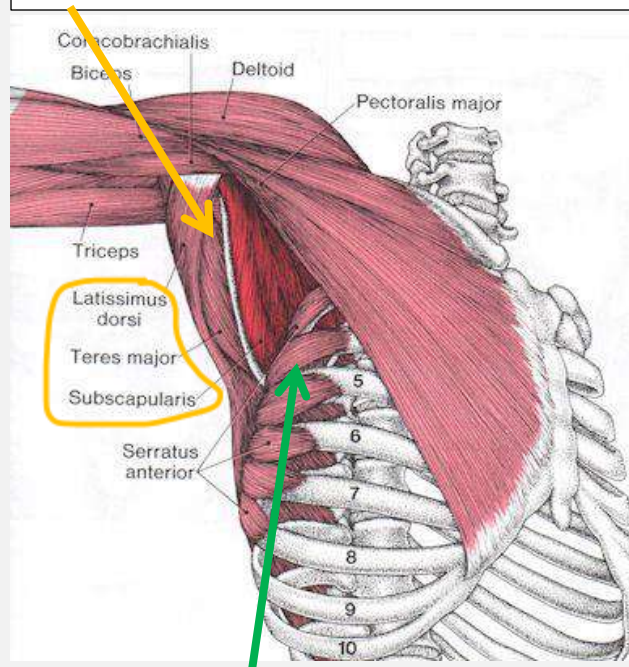


Pectoralis Major

Subclavius

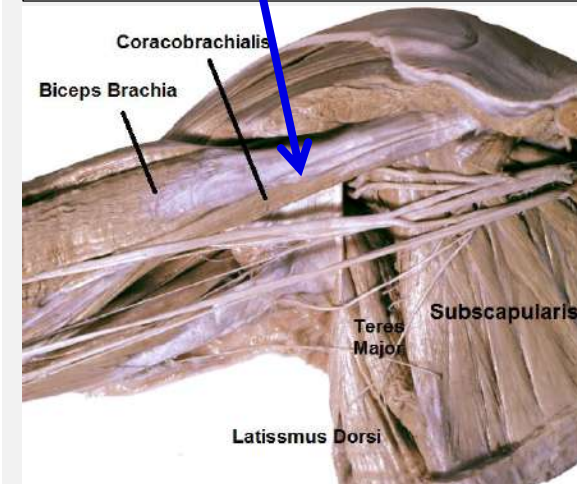
Pectoralis Minor

Posterior Wall:



Medial Wall:

Lateral Wall:



Walls of the Axilla

4 walls made up as follows: (each wall has 3 structures)

Anterior Wall:

- pectoralis major,
- pectoralis minor,
- subclavius muscles

Lateral Wall:

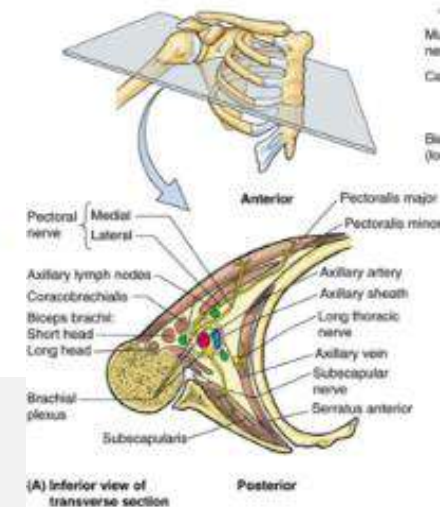
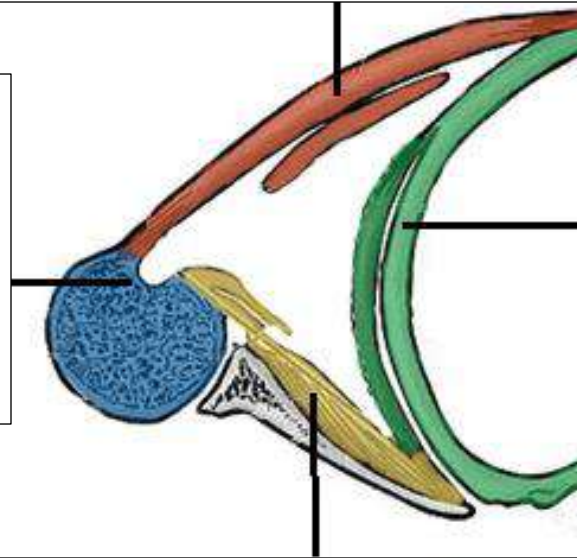
- coracobrachialis &
- biceps muscles
- bicipital groove of humerus

Medial Wall:

- upper 4-5 ribs &
- inter- costal spaces
- covered by serratus anterior muscle.

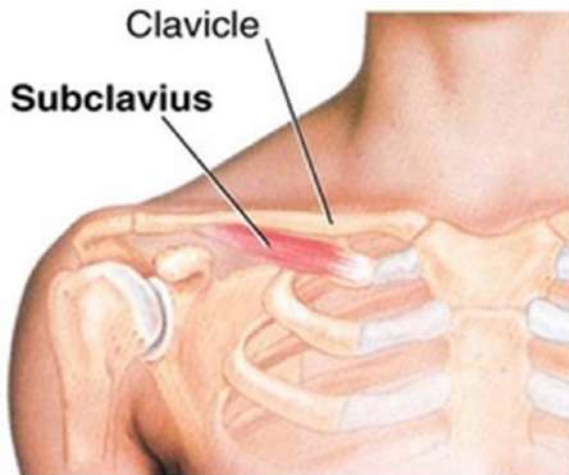
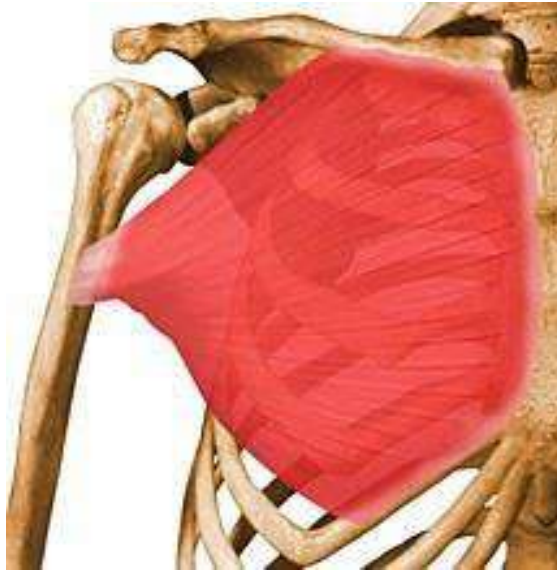
Posterior Wall:

- subscapularis,
- latissimus dorsi,
- teres major muscles

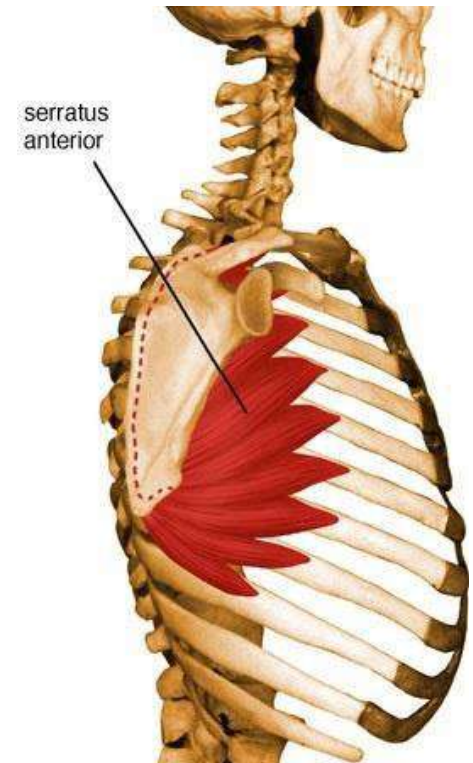


- **Muscles of anterior – medial wall:**

- Pectoralis major
- Pectoralis minor
- Subclavius
- Serratus anterior

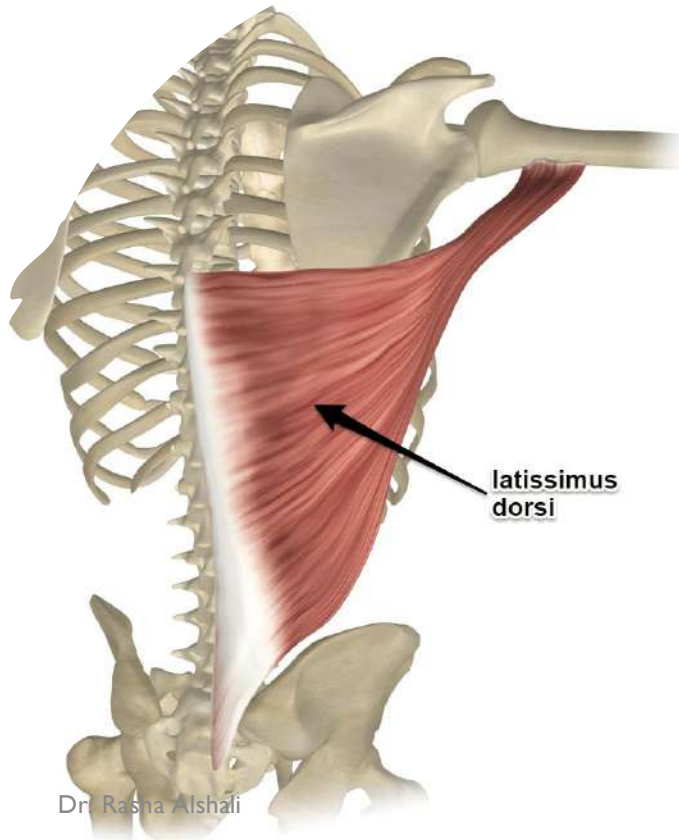


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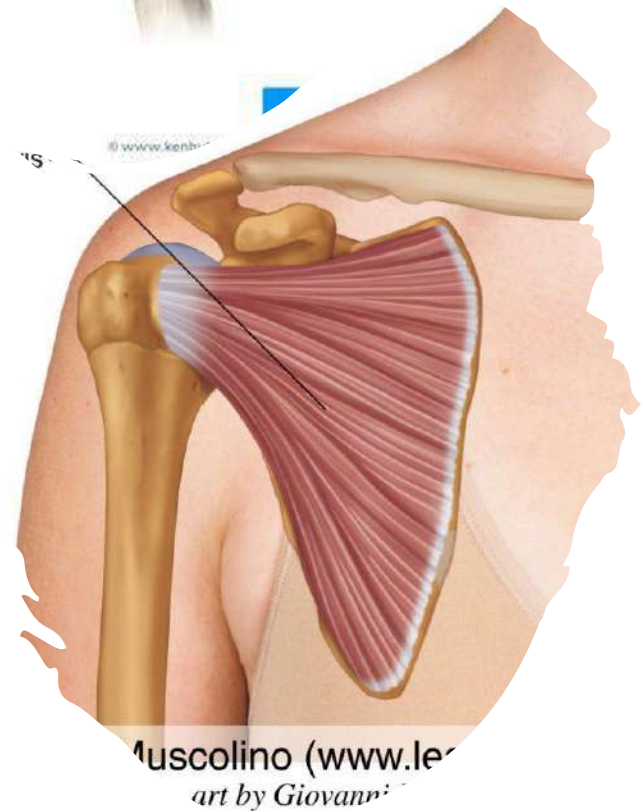


- **Muscles of posterior wall:**

- Subscapularis
- Latissimus dorsi
- Teres major



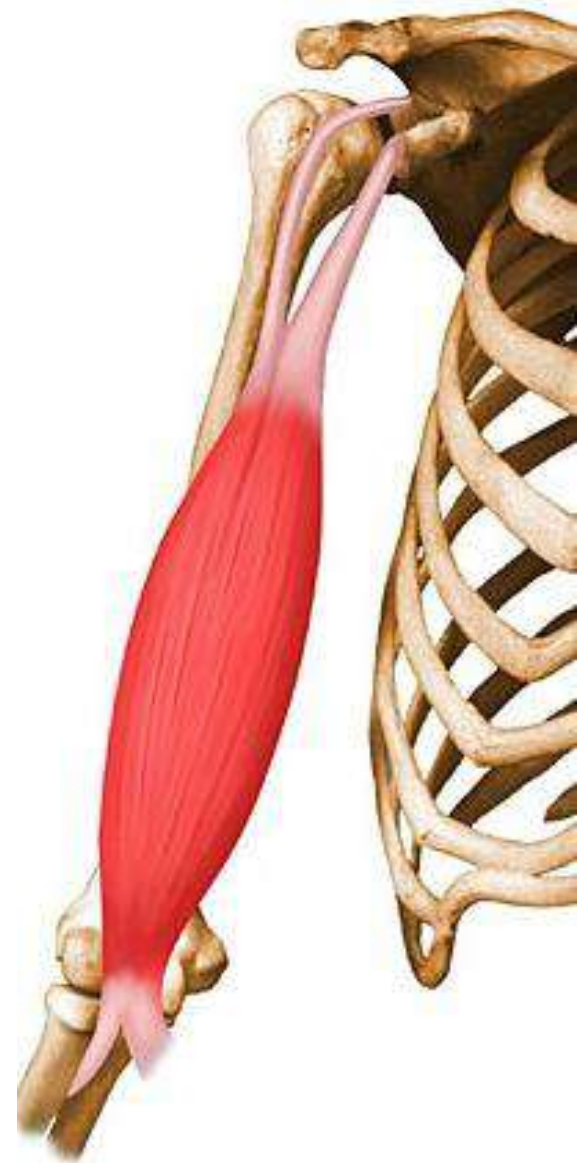
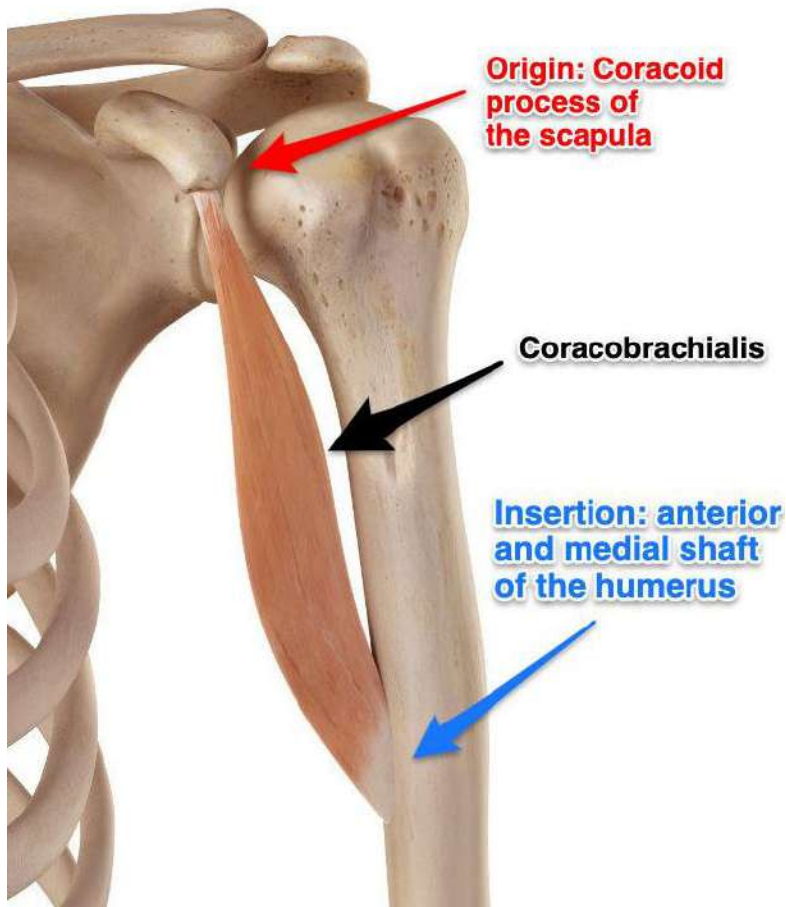
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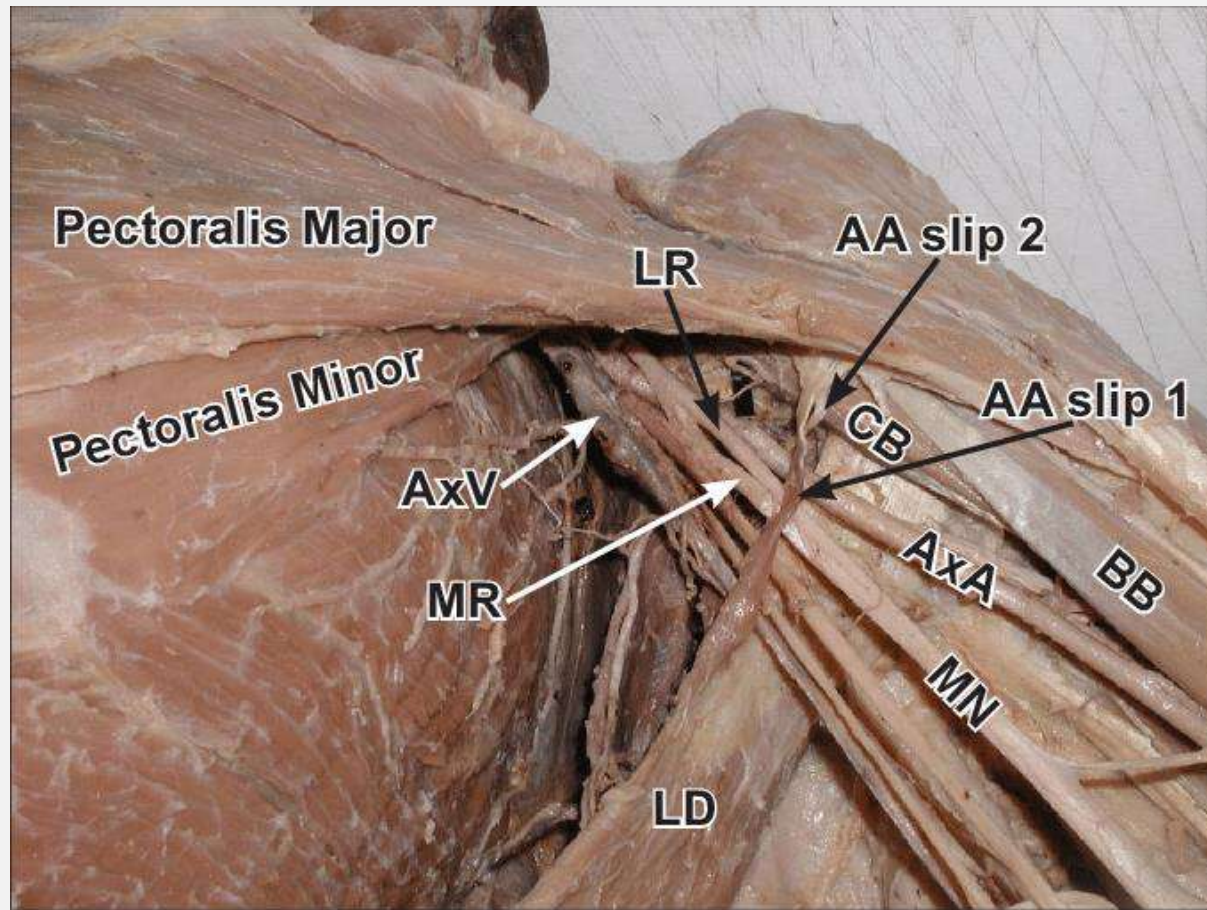
Muscolino (www.le
art by Giovanni

Muscles of lateral wall:

- Coracobrachialis
- Biceps brachii



Contents of the Axilla



<https://www.youtube.com/watch?v=NEJenl3zhio>

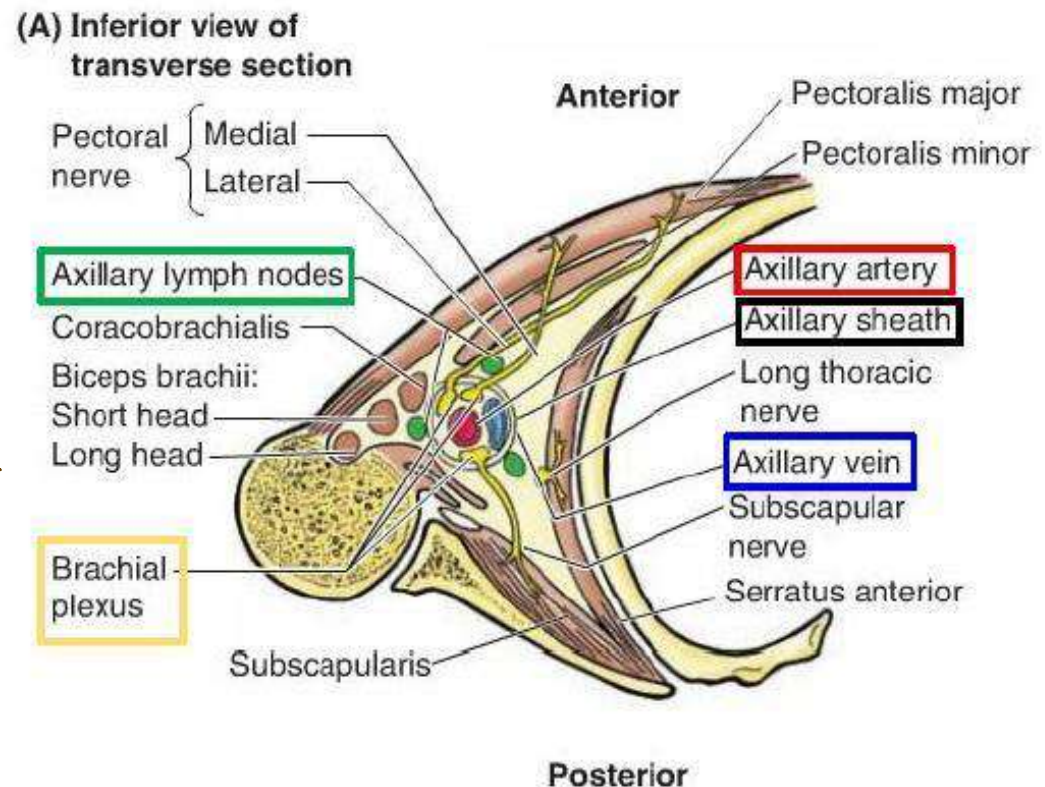
Min 0- 5

Contents of the Axilla

These structures are embedded in fat:

- The **axillary artery** and its branches,
- The **axillary vein** and its tributaries,
- The **brachial plexus**, cords and branches,
- The **Axillary lymph nodes & vessels.(except infraclavicular L.N.)**
- The **axillary tail** of the breast
- The axillary fat

Which contents are enclosed in the axillary sheath.



<https://www.youtube.com/watch?v=NEJenI3zhio>

Min 0- 5

THE AXILLARY ARTERY

Begins: at 'lateral border of **1st rib** as a continuation of **subclavian artery**

Ends: at 'lower border of '**teres major** muscle, to continue as **brachial artery**

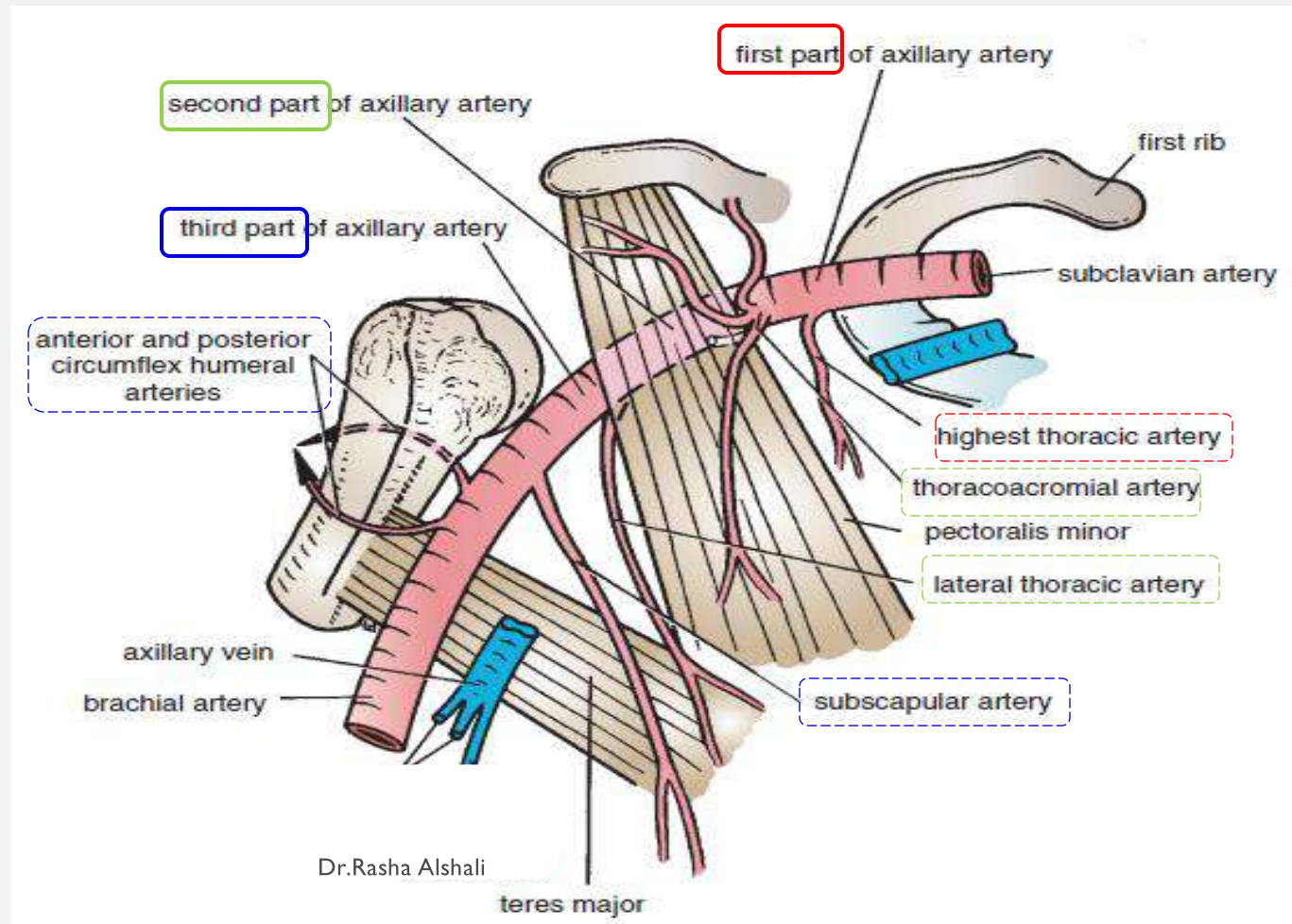
The **pectoralis minor muscle** crosses in front of 'axillary artery and divides it into **three parts**.

1st → **1 branch**

2nd → **2 branches**

3rd → **3 branches**

Begins
Ends
Parts
Branches
Relations



The Axillary Artery - Parts & Branches

*The largest
branch*

First Part:

Extends from the lateral/ outer border of 1st rib
to the upper/ medial border of pectoralis **minor**.

Branches: 1- The **highest thoracic artery** (runs along the upper border of the pectoralis minor).

Second Part:

This lies **behind** the pectoralis minor muscle.

Branches: 1- The **thoracoacromial artery**.

2- The **lateral thoracic artery**

Third Part:

Extends from the lower/ lateral border of pectoralis **minor**
to the lower / lateral border of teres **major**.

Branches: 1- The **subscapular artery** (runs along the lower border of subscapularis muscle).

2- The **anterior circumflex humeral artery**

3- The **posterior circumflex humeral artery**

(2 & 3 wind around the front & back of 'surgical neck of the humerus, respectively).

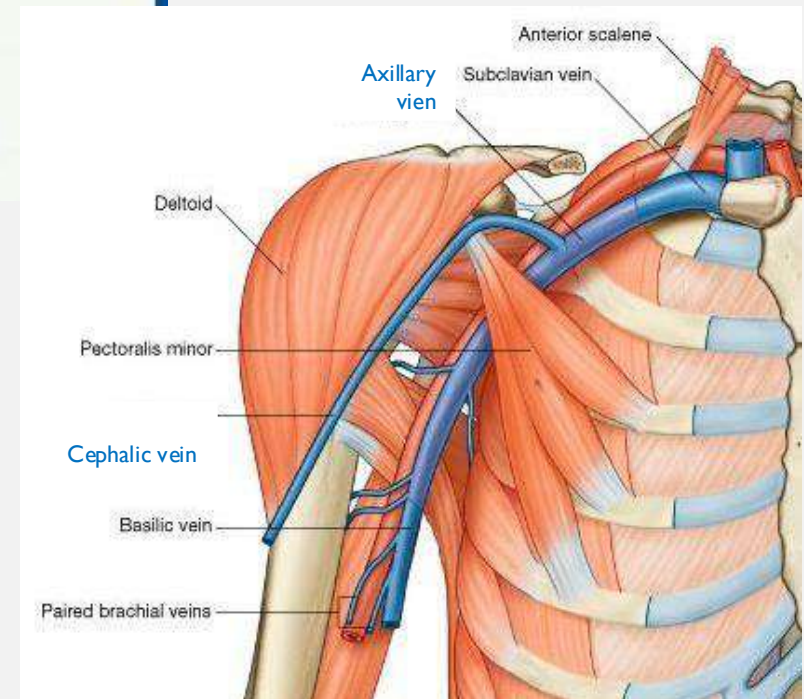
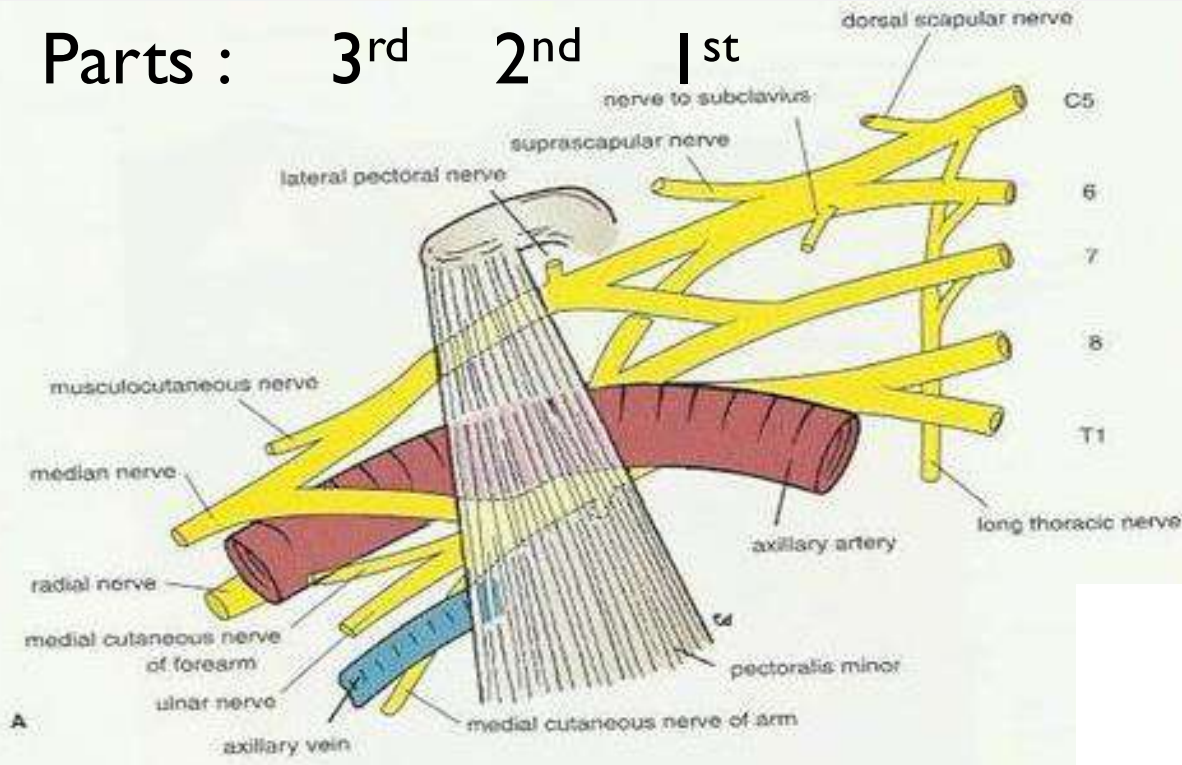
The Axillary Artery - Relations

Parts :

3rd

2nd

1st



- The **axillary artery** runs lateral to the **axillary vein**.
- The 1st part of **axillary artery** is Crossed anteriorly by **cephalic vein**.

AXILLARY VEIN

Begins: at the lower border of the **teres major** muscle.

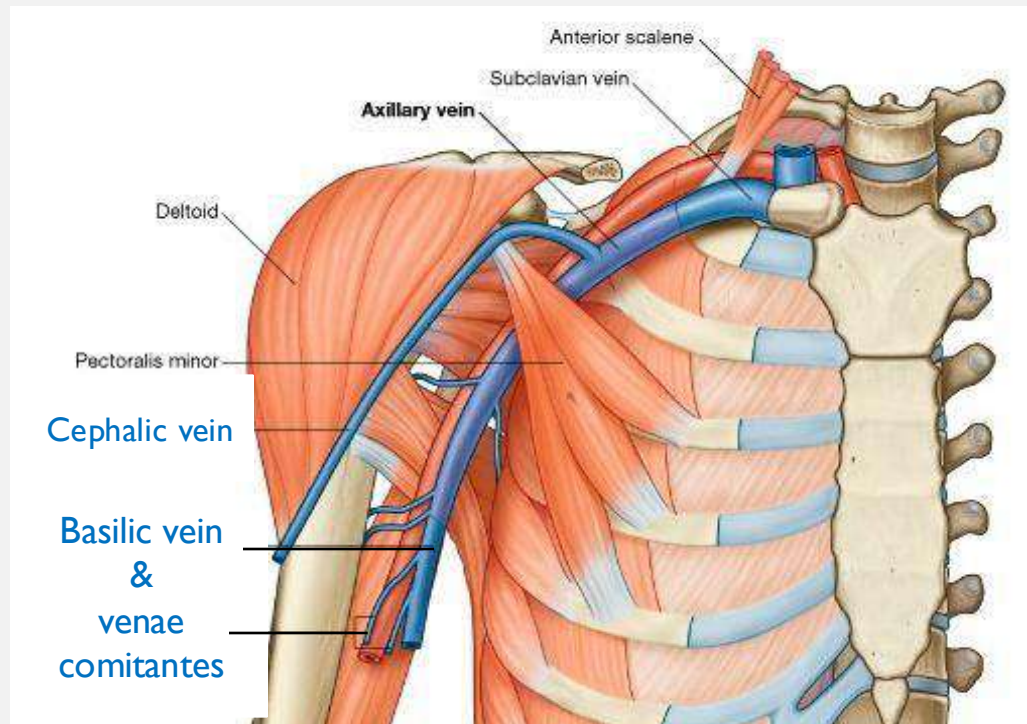
Formed by: the union of 'venae comitantes of the brachial artery & the basilic vein.

- It runs upward **medial** to the axillary artery.

Ends: at the lateral border of 1st rib by becoming **subclavian vein**.

Tributaries:

- correspond to the branches of the axillary artery,
- the cephalic vein.

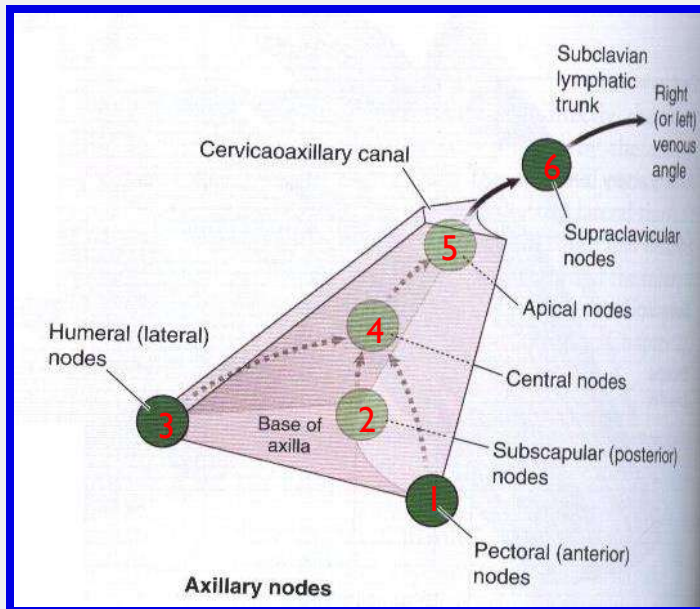
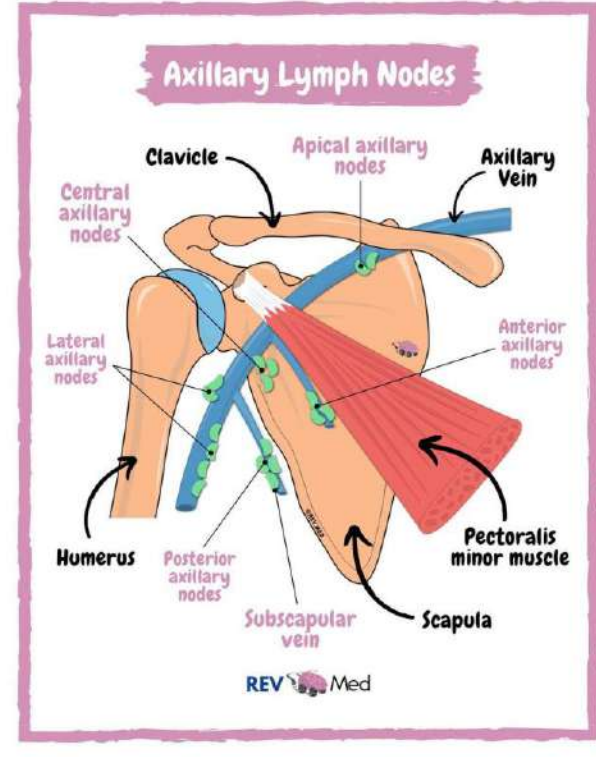


Axillary Lymph Nodes

**They are (20 to 30 in number)
arranged in five groups:**

- 1) **Anterior** (pectoral)
- 2) **Posterior** (subscapular)
- 3) **Lateral** (humeral)
- 4) **Central**: near 'base of axilla
- 5) **Apical**: near 'apex of axilla

Infraclavicular (deltopectoral)



They drain the lymph from:

1. The **upper limb**.
2. **Thoracic and abdominal wall** till the level of the **umbilicus**.
3. Lateral quadrants of the **breast**.

Axillary Lymph Nodes

Anterior (pectoral) group: Lying along lower border of pectoralis minor behind pectoralis major, Receive lymphatics from lateral quadrants of the breast and superficially from the anterolateral abdominal wall above level of umbilicus.

Posterior (subscapular) group: Lying in front of subscapularis, Receive superficial lymphatics from the back, down as far as level of iliac crests.

Lateral group: Lying along the medial side of the axillary vein near humerus, Receive most of the lymphatics of the upper limb (except those superficial draining the lateral side to infraclavicular nodes).

Central group: Lying in the center of axilla in the axillary fat, Receive lymphatics from the above three groups.

• **Apical group:** Lying at the apex of axilla at lateral border of the 1st rib, Receive lymphatics from all the other axillary nodes.

• **Infraclavicular (deltopectoral) group:**

These nodes are **not strictly axillary nodes** because they are located outside the axilla. They lie in the groove between the deltoid and pectoralis major muscles. Receive superficial lymphatics from the lateral side of the hand, forearm, and arm.

Axillary Lymph Nodes

- The apical nodes ...

drain into the **subclavian lymph trunk**.

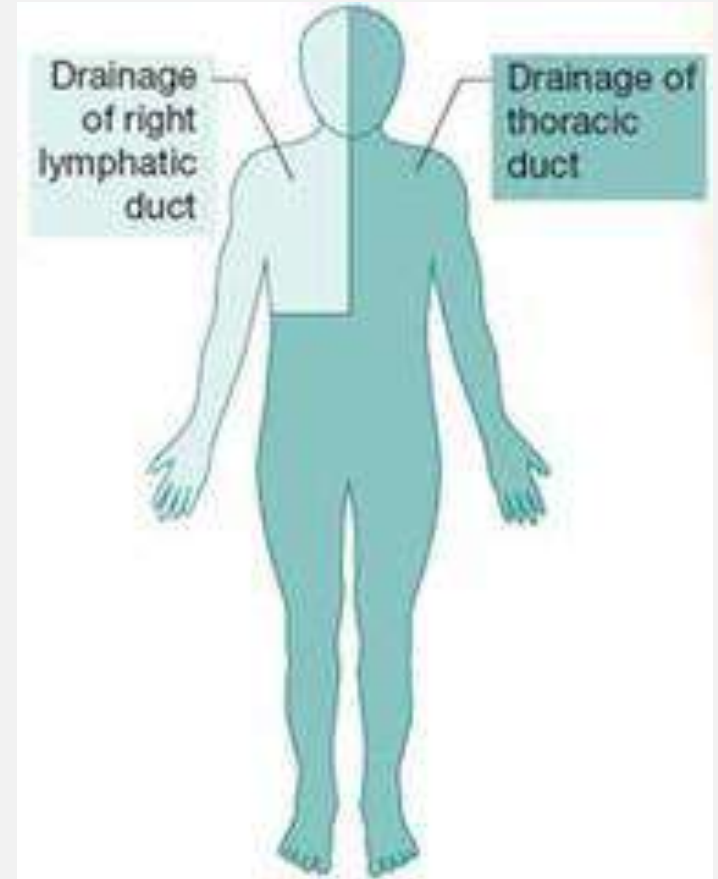
- On the left side:

this trunk drains into the **thoracic duct**;

- On the right side:

it drains into the **right lymph trunk**.

Alternatively, lymph trunks may drain directly into one of the large veins at the root of ' neck.



CLINICAL POINTS

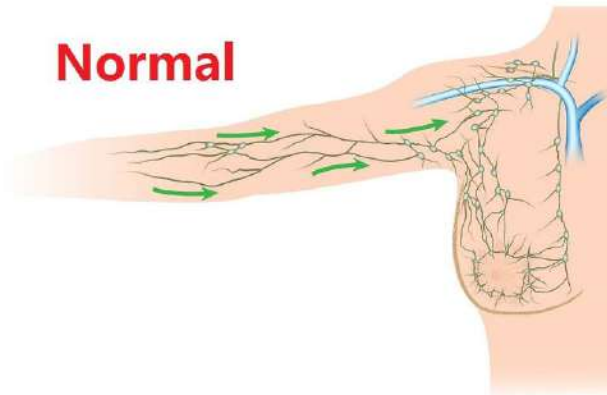
Upper limb lymphoedema

HOW ?

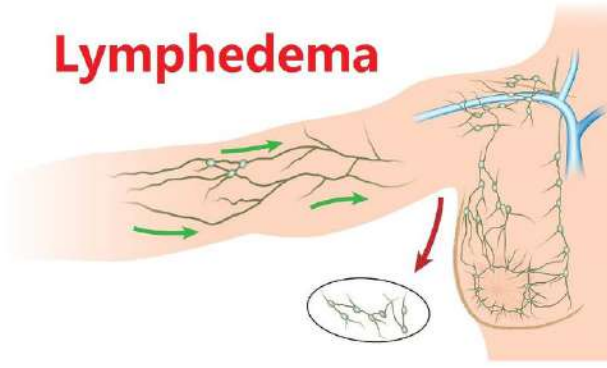
Why?



Normal



Lymphedema



Lymphedema From Breast Cancer



BRACHIAL PLEXUS

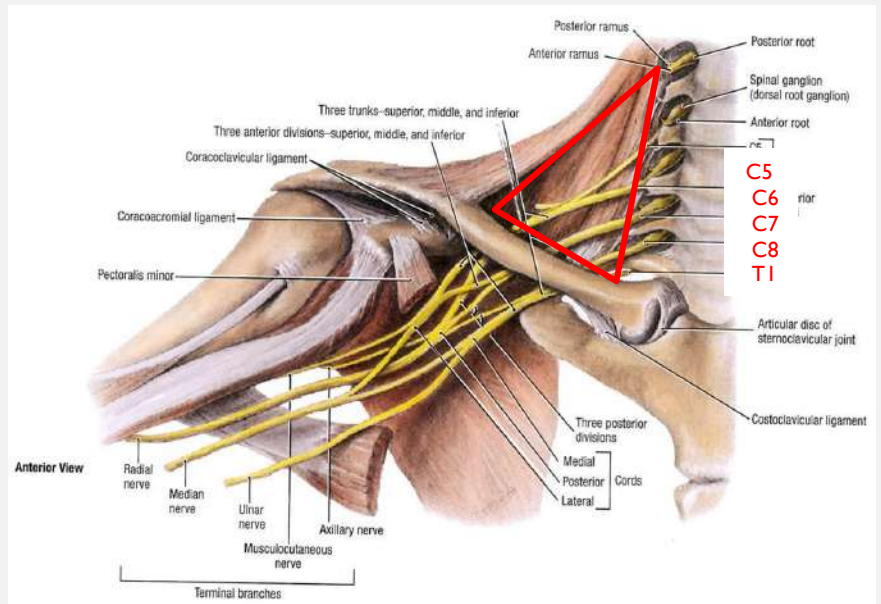
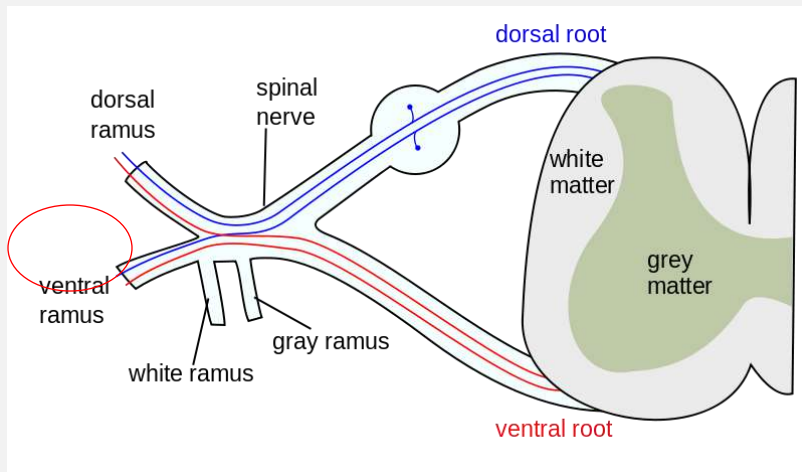
- It is a plexus of nerves entering the upper limb to provide:
 - **Sensory** innervations (to skin and joints)
 - **Motor** innervations (to the muscles)

Site of formation of the brachial plexus:

At the root of the neck, (posterior triangle of the neck)

Formation:

by the union of **anterior rami** of spinal nerves: **C5, C6, C7, C8 and T1**
the nerves form a complicated plexus (**the brachial plexus**)



Brachial Plexus

Lies in:

Upper limb

Axilla

Posterior triangle of 'neck

Branches

Cords

Divisions
Anterior or Posterior

Trunks

Roots

C5

C6

C7

C8

T1

Musculocutaneous

Lateral

Upper

Axillary

Median

Posterior

Middle

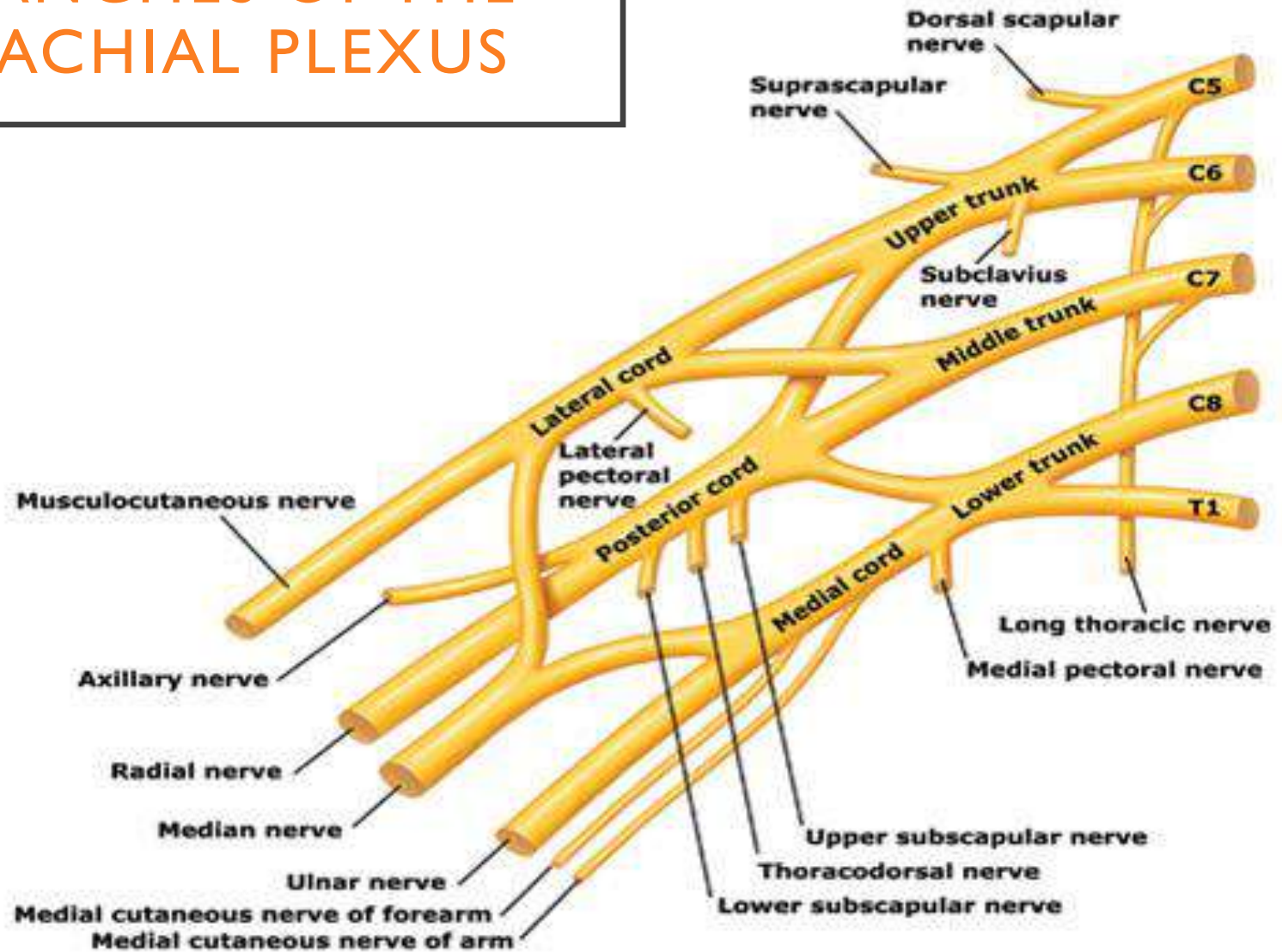
Radial

Ulnar

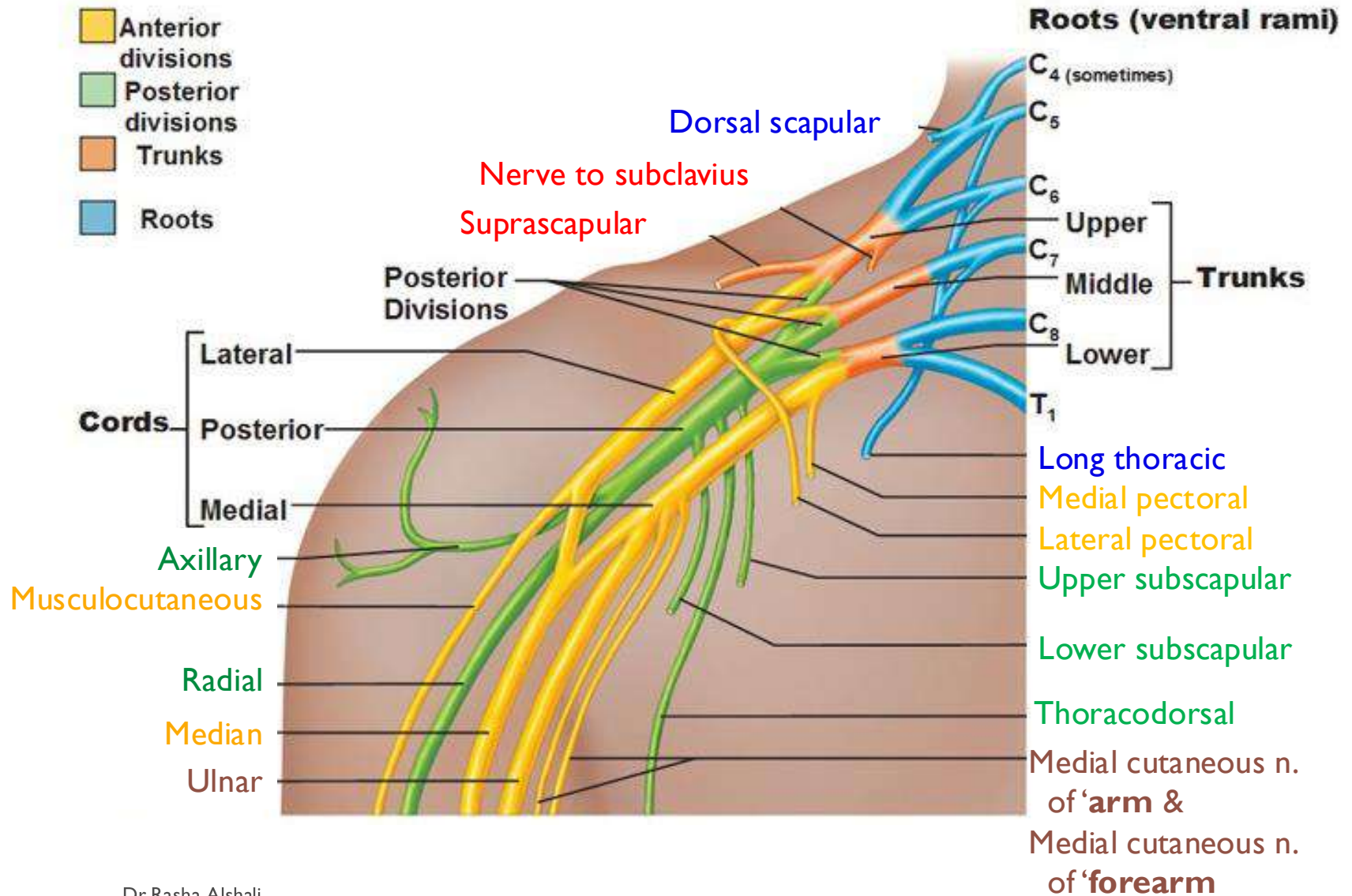
Medial

Lower

BRANCHES OF THE BRACHIAL PLEXUS



BRANCHES OF THE BRACHIAL PLEXUS



Branches of the Brachial Plexus

■ Roots (2)

Dorsal scapular nerve (C5)

Long thoracic nerve (C5, 6, and 7)

■ Upper trunk (2)

Nerve to subclavius (C5 and 6)

Suprascapular nerve

■ Lateral cord (3)

Lateral pectoral nerve

Lateral root of median nerve

Musculocutaneous nerve

■ Medial cord (5)

Medial pectoral nerve

Medial root of median nerve

Medial cutaneous nerve of arm

Medial cutaneous nerve of forearm

Ulnar nerve

■ Posterior cord (5)

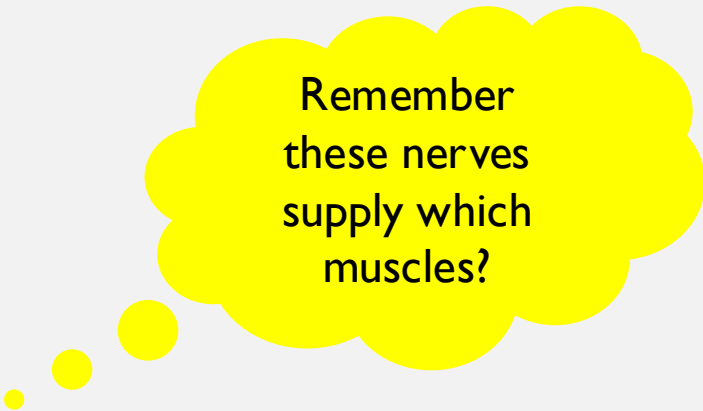
Upper subscapular nerve

Lower subscapular nerve

Thoracodorsal nerve

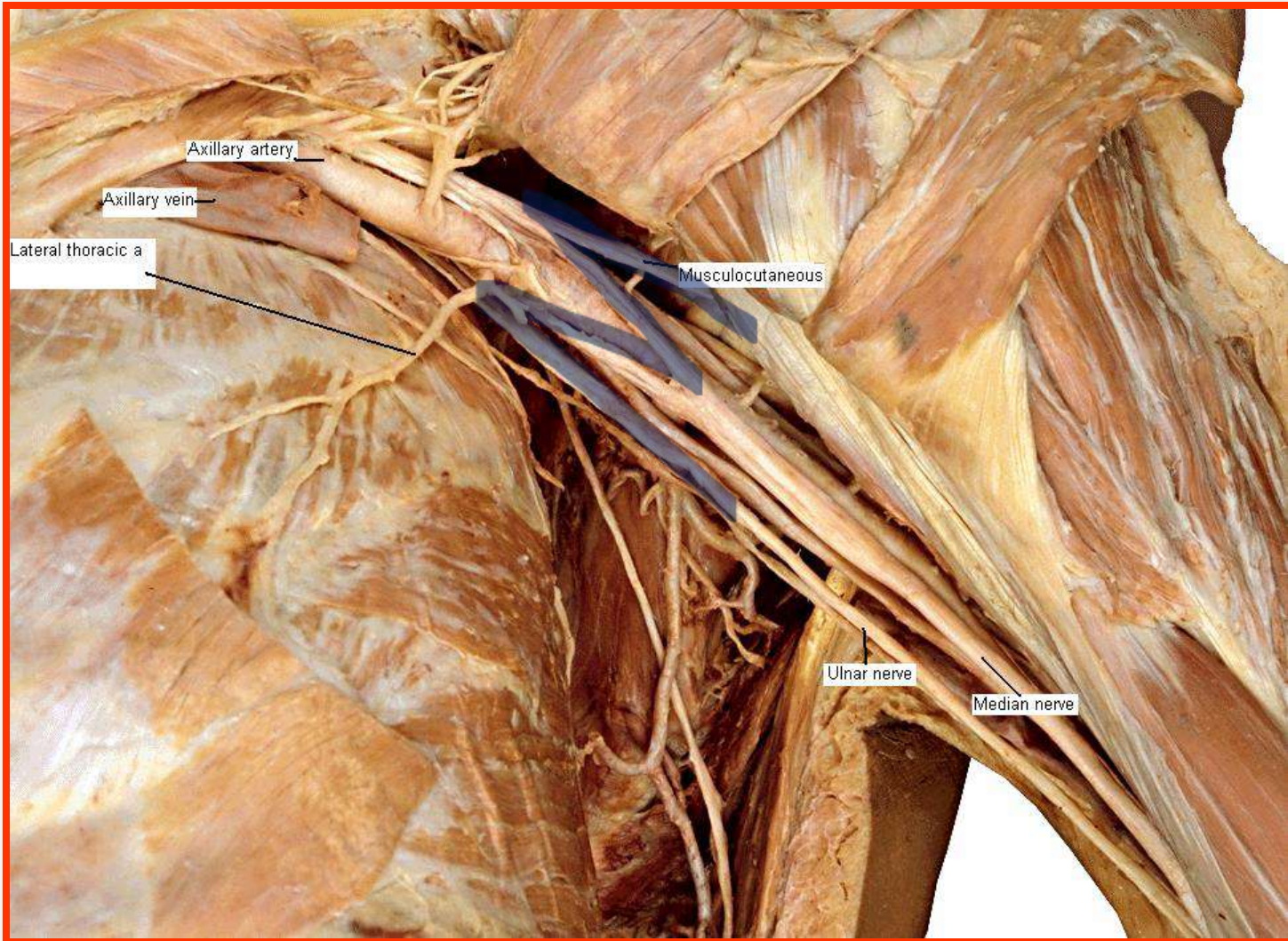
Axillary nerve

Radial nerve



Remember
these nerves
supply which
muscles?

The axillary artery & Brachial Plexus





Dr. Rasha Alshali

Associate Professor

Anatomy Department, Faculty of Medicine, KAU

PhD (The University of Liverpool, UK)

✉ rmalshali@kau.edu.sa

☎. Ext: 23040

